

Md. Abu Sayed Chowdhury

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EDUCATION

Bangladesh University of Engineering and Technology (BUET) 2025–Present

M.Sc. in Electrical and Electronic Engineering, in progress

Focus: VLSI design and memristor-based computing

Bangladesh University of Engineering and Technology (BUET) 2020–2025

B.Sc. in Electrical and Electronic Engineering

GPA: 3.95/4.00; Class rank: 7/213. Dean's List in all academic years.

Relevant coursework: VLSI Circuits and Design; Analog Integrated Circuits; Semiconductor and Nano Devices; Introduction to Nanotechnology and Nanoelectronics; Processing and Fabrication Technology; Privacy Preserving Machine Learning; Spintronics.

RESEARCH INTERESTS

- **In-memory computing:** Memristor/ReRAM crossbars and hybrid memristor–CMOS circuits that compute Boolean and arithmetic functions where the data is stored, cutting data movement.
- **Neuromorphic hardware:** Spiking neural hardware, memristor synapses, and low-latency accelerators for edge AI.
- **Low-power/mixed-signal VLSI:** Subthreshold circuits, power gating, DVFS, and precision ADC/DACs.

PUBLICATIONS & PRESENTATIONS

M. A. S. Chowdhury and O. Hassan, “In-Memory Boolean Logic Computation in Binary Neural Network Inspired Memristor Crossbar Array,” in *Proc. 10th IEEE Electron Devices Technology & Manufacturing Conference (EDTM)*, Penang, Malaysia, 2026, pp. 1–3.

DOI: 10.1109/EDTM65772.2026.11497945 | IEEE Xplore | *Poster presenter, March 3, 2026*

RESEARCH EXPERIENCE

Undergraduate Thesis Research, BUET Jan 2024–Mar 2025

In-Memory Logic Implementation in Memristor-Based Crossbar Arrays

Supervisor: Dr. Orchi Hassan

- Developed an in-memory Boolean logic scheme modeled on binary neural networks; each binary weight is stored in two memristors in a crossbar array.
- Designed the CMOS programming and readout circuits in Cadence Virtuoso (GPDK045). SPICE simulations showed NAND and XOR gates and a cascaded full adder.
- Wrote and calibrated Verilog-A models based on VTEAM (threshold switching, nonlinear current–voltage behavior) for CMOS–memristor co-simulation.

ACADEMIC APPOINTMENTS & TEACHING

Lecturer, BRAC University Jan 2026–Present

Department of Electrical and Electronic Engineering, BSRM School of Engineering

- Teaching Signals and Systems (EEE243), Numerical Techniques (EEE282), and Modeling and Simulation (EEE382).

Adjunct Lecturer, BUET Apr 2025–Jan 2026

Department of Electrical and Electronic Engineering

- Ran lab sections for Electrical Circuits, Numerical Techniques, and Digital Signal Processing: pre-lab briefings, demonstrations, and help debugging experiments.
- Helped update the lab curriculum and assessments. Gave students feedback and mentoring.

SELECTED ENGINEERING PROJECTS

16-bit Carry Skip Adder: RTL-to-GDSII Design

Dec 2024

SystemVerilog, Cadence Genus/Innovus, GPDK045 | GitHub

- Designed a block-based 16-bit carry skip adder in SystemVerilog, with propagate/generate logic and 2:1 multiplexers that skip ripple-carry blocks.
- Wrote a layered SystemVerilog testbench with constrained-random stimulus. Functional coverage reached 100% on eight signed-arithmetic flag scenarios (Negative, Zero, Carry, Overflow).
- Synthesized block sizes $k = 2, 4, 8$ and compared power, delay, and area. Picked $k = 4$: 1.24 ns critical-path delay, 3.88 μW total power, 136.8 μm^2 cell area.
- Took the design through floorplanning, power mesh, placement, clock-tree synthesis, and routing in Innovus. The exported GDSII passed DRC/LVS with no setup or hold violations.

Analog 8:1 Multiplexer for Biomedical Signal Acquisition

Feb 2024

Cadence Virtuoso, GPDK045, CMOS transmission gates | GitHub

- Designed an 8:1 analog multiplexer for ECG/EEG signal acquisition: 24 CMOS transmission gates, eight input channels, three select lines.
- Characterized transient, AC, DC, and noise behavior at a 3 V supply: 71.8 μW power, 1.576 pC charge injection, 92.66 ns switching delay, 3.5 MHz –3 dB bandwidth, and about 850 Ω on-resistance.
- Sized the transistors against GPDK045 models to reduce crosstalk and signal distortion.

FPGA-Based Intelligent Home Automation

Aug 2023

Verilog, Intel Quartus, Altera Flex EPF10K70 | GitHub

- Wrote Verilog state machines that track people entering and leaving and switch the lights on when the room is occupied; the lights shut off automatically after 10 seconds.
- Used two infrared sensors for occupancy and NTC thermistors for temperature; the fan turns on above 30 °C.
- Built four-speed PWM fan control with duty cycle set by occupancy: 1–3 people at 25%, 4–5 at 50%, 6–7 at 75%, 8–9 at 100%.

TECHNICAL SKILLS

- IC design and EDA:** Cadence Virtuoso (schematic/layout), Genus (synthesis), Innovus (place-and-route, clock-tree synthesis); GPDK045; DRC/LVS verification.
- Hardware description and modeling:** Verilog, SystemVerilog, Verilog-A; VTEAM memristor modeling; RTL verification, functional coverage, finite-state machines, and FPGA development with Intel Quartus Prime.
- Simulation and programming:** SPICE-based circuit simulation, LTspice, PSpice, Proteus; Python (NumPy, pandas, PyTorch), MATLAB/Simulink, C/C++, ARM assembly.
- Embedded platforms:** STM32, Arduino, ESP8266/ESP32, and Altera FPGAs.

HONORS & AWARDS

- First Runner-Up, VLSITHON 2024:** University-wide VLSI design competition organized by Neural Semiconductor with BUET EEE Day 2024.
- University Merit Scholarship, BUET:** Awarded for five semesters.

ACADEMIC REFERENCES

Dr. Orchi Hassan

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Associate Professor, Department of Electrical and Electronic Engineering, BUET

Undergraduate thesis supervisor

Dr. Md. Shah Alam

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Professor, Department of Electrical and Electronic Engineering, BUET

Undergraduate academic advisor